

# Integration Pulse

## Operations, User, and Technical Administration Guide

Version 0.1 Draft

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Prepared for: HR Operations, HRIS, Integration Support, SAP BTP Administrators, and Application Owners

**Confidentiality:** Internal / project use. Do not include secrets, tenant URLs, or production payload data.

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# Document Control

| Field          | Value                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------|
| Document title | Integration Pulse - Operations, User, and Technical Administration Guide                                   |
| Version        | 0.1 Draft                                                                                                  |
| Status         | Draft for review                                                                                           |
| Prepared date  | 2026-06-22                                                                                                 |
| Prepared for   | HR operations, HRIS, integration support, SAP BTP administrators, and application owners                   |
| Owner          | Application owner to be confirmed                                                                          |
| Review cadence | Review before each production release and after significant Integration Suite or BTP configuration changes |

| Field           | Value                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Related systems | SAP BTP Integration Suite, SAPUI5 frontend, optional FastAPI proxy, PostgreSQL payload storage, SAP BTP destination routing |

## Intended audience

This guide is written for a mixed audience. HR operations users can use it to understand day-to-day monitoring and escalation workflows. HRIS administrators and integration support teams can use it to review externalized parameters, schedules, run history, and payload summaries. SAP Integration Suite developers and SAP BTP administrators can use the technical sections to understand API routing, payload storage, deployment assumptions, and open implementation items.

## Environment note

Features may vary by environment, deployment mode, and permissions. The current codebase includes mock mode, BTP destination mode, and optional FastAPI proxy mode. The final production entry point, authentication model, and role enforcement must be validated for the target SAP BTP environment.

## Version history

| Version | Date       | Author                                     | Summary                                                      |
|---------|------------|--------------------------------------------|--------------------------------------------------------------|
| 0.1     | 2026-06-22 | Generated from current repository evidence | Initial operations, user, and technical administration guide |

## Executive Overview

Integration Pulse is an operations application for deployed SAP BTP Integration Suite integrations. It gives HR operations and integration support teams a cleaner way to view deployed integrations, understand which Source System and Target System each integration supports, review configuration values, redeploy supported integrations, and investigate monitoring activity.

SAP Integration Suite is powerful, but routine HR operations work can be difficult for nontechnical users because important information is spread across runtime artifacts, design-time artifacts, externalized parameters, deployment actions, and Message Processing Logs. Integration Pulse presents that information in a guided business-readable interface.

The application helps answer practical questions:

1. Which integrations are deployed?
2. Which Source Systems or Target Systems are affected?
3. Which integrations failed or have warnings?
4. What configuration may need adjustment?
5. What schedule is configured for a timer-driven integration?
6. What summary result data was sent for a run?
7. Which vendor, HR platform, payroll system, or third-party service is impacted?

## What Integration Pulse is and is not

| Integration Pulse is                                                                | Integration Pulse is not                                                   |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| A SAPUI5 operations console for deployed Integration Suite runtime artifacts        | A full Integration Suite design-time replacement                           |
| A business-readable monitoring and configuration interface                          | A data warehouse for employee, payroll, benefits, or transactional records |
| A tool for reviewing externalized parameters and redeploying supported integrations | A source-code editor for iFlows                                            |
| A way to capture integration-run payload summaries for support review               | A place to store detailed PII or full transactional payloads               |
| A frontend with optional FastAPI proxy and PostgreSQL-backed payload storage        | A fully implemented enterprise role model in the current codebase          |

## Current implementation summary

The repository implements a SAPUI5 XML/MVC frontend with routes for Home, Integrations, Integration Detail, Monitoring, and Monitoring Detail. It includes BTP destination live mode for Integration Suite OData calls, an optional FastAPI proxy backend, and PostgreSQL-backed payload capture. Authorization roles are planned but not enforced in code.

## Product Capabilities At A Glance

| Capability                    | Business purpose                            | Audience              | Status      | Application area          | Notes                                                               |
|-------------------------------|---------------------------------------------|-----------------------|-------------|---------------------------|---------------------------------------------------------------------|
| Runtime artifact discovery    | Show deployed integrations                  | HR Ops, HRIS, support | Implemented | Integrations              | Filters deployed/runtime-like artifacts in frontend                 |
| Source System grouping        | Organize integrations by originating system | HR Ops, support       | Implemented | Integrations, Monitoring  | Uses Sender metadata where available                                |
| Target System grouping        | Organize integrations by receiving system   | HR Ops, support       | Implemented | Integrations, Monitoring  | Uses Receiver metadata where available                              |
| Search integrations           | Find integrations quickly                   | All users             | Implemented | Integrations              | Searches names, IDs, packages, and system labels                    |
| Externalized parameter review | Review configurable values                  | HRIS, support         | Implemented | Integration Detail        | Values are loaded from IntegrationDesignTimeArtifactsConfigurations |
| Configuration editing         | Update supported externalized values        | HRIS, support         | Implemented | Integration Detail        | No role enforcement currently verified                              |
| Redeployment                  | Redeploy after changes                      | Integration support   | Implemented | Integration Detail, cards | Confirmation dialog is present                                      |

| Capability            | Business purpose                          | Audience        | Status                  | Application area                  | Notes                                                           |
|-----------------------|-------------------------------------------|-----------------|-------------------------|-----------------------------------|-----------------------------------------------------------------|
| Schedule management   | Make timer values easier to edit          | HRIS, support   | Implemented             | Integration Detail                | Converts UI selections to cron-like expression strings          |
| Monitoring overview   | Show system-level health                  | HR Ops, support | Implemented             | Monitoring                        | Passed, Failed, Warning counts by Source/Target System          |
| Run history           | Review recent executions                  | Support         | Implemented             | Monitoring Detail                 | Message logs loaded by integration ID                           |
| Payload capture       | Store run result payload summaries        | Support, HRIS   | Implemented backend/API | Payload API                       | PostgreSQL required outside mock mode                           |
| Payload preview       | Preview small JSON/CSV/XML/text summaries | Support         | Implemented             | Monitoring Detail, payload dialog | Payloads over 100 KB are download-only                          |
| Retention             | Remove old payloads                       | Platform owners | Implemented on access   | Backend payload storage           | Expired rows pruned when payload storage functions run          |
| Roles and permissions | Restrict view/edit actions                | App owners      | Planned                 | Security                          | Read Only and Editable are not enforced in the current codebase |

# Roles And Responsibilities

## Current verified authorization behavior

The current repository does not contain an enforced Read Only or Editable role model in the SAPUI5 application, FastAPI routes, XSUAA configuration, approuter configuration, or deployment descriptor. The UI exposes edit and deploy controls based on application flow, not verified role checks.

This means the role model below is a planned/proposed model and must be implemented through the target identity provider, BTP security configuration, backend authorization checks, frontend control visibility, or a combination of those controls.

## Planned / Proposed Role Model

| Planned role | Intended access                                                                                         | Enforcement status                         |
|--------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Read Only    | View integrations, Source/Target grouping, monitoring, run history, logs, and permitted payload details | Planned - not enforced in current codebase |
| Editable     | All Read Only activities plus edit supported configuration values and redeploy integrations             | Planned - not enforced in current codebase |

## Responsibility matrix

| Group                  | Responsibilities                                                                                                                          |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| HR Operations          | Monitor integration health, review business-readable statuses, escalate failures, avoid downloading or sharing restricted payload content |
| HRIS Administrators    | Review parameters, understand schedule behavior, coordinate approved changes, validate run outcomes                                       |
| Integration Support    | Investigate failed or warning runs, review Message Processing Logs, troubleshoot connectivity or payload capture issues                   |
| Integration Developers | Maintain iFlows, configure payload summary calls, ensure payloads do not include prohibited data, support deployment issues               |
| SAP BTP Administrators | Maintain destinations, service bindings, environment variables, PostgreSQL access, deployment routes, and production security controls    |
| Application Owners     | Own release readiness, role model, documentation review, support model, and privacy guardrails                                            |

## Future role expansion placeholder

Future roles may include Payload Viewer, Configuration Editor, Deployment Operator, Administrator, and Audit Viewer. These must be validated against the production authorization architecture before documentation can describe them as enforced.



# Getting Started

## Business explanation

Integration Pulse is intended to be opened by users who need to operate or monitor deployed integrations without working directly in low-level Integration Suite screens. The application navigation includes Home, Integrations, and Monitoring.

## Before you begin

- Confirm the application URL for the target environment.
- Confirm whether the environment is mock, BTP destination, or proxy mode.
- Confirm that the backend payload API is running if payload Results are expected.
- Confirm that PostgreSQL payload storage is configured for non-mock payload capture.
- Confirm whether the organization has implemented role restrictions outside the current repository.

## Access and environment behavior verified in code

The SAPUI5 frontend reads runtime settings from `webapp/service/config.js`. The current code defaults to `useMock: false`, `liveMode: destination`, `destinationBaseUrl: /api/v1`, `payloadBaseUrl: /payload-api/v1`, and `backendBaseUrl: http://localhost:8000`.

Runtime URL overrides are supported:

| URL parameter               | Effect                                |
|-----------------------------|---------------------------------------|
| ?mock=true                  | Uses local mock JSON fixtures         |
| ?mock=false                 | Uses configured live mode             |
| ?mock=false&api=proxy       | Calls the optional FastAPI proxy      |
| ?mock=false&api=destination | Calls same-origin destination routing |

## Supported browser assumptions

The manifest declares desktop, tablet, and phone device support. The application is built with OpenUI5 1.120.20 and SAPUI5-style XML/MVC views.

# User Interface Overview

## Home

The Home page contains tiles for Integrations and Monitoring. The shell sidebar groups Integrations and Monitoring beneath Home. This provides a simple landing point for HR operations users.

**Screenshot needed: Home page showing Integrations and Monitoring tiles.**

## Integrations

The Integrations page displays deployed integration flows. Users can switch between Source System and Target System grouping. Category tiles show the system name and integration count. Selecting a category shows the integrations for that system without leaving the application shell.

**Screenshot needed: Integrations page grouped by Source System.**

## Integration Detail

The Integration Detail page shows metadata and externalized parameters. Parameters are grouped by prefix when possible. Timer-like parameters are rendered with a scheduling UI that supports Run Once, Schedule on Day, Schedule to Recur, and Advanced modes. Deploy actions are guarded by confirmation dialogs.

**Screenshot needed: Integration Detail externalized parameters and schedule controls.**

## Monitoring

The Monitoring page summarizes integration health by Source System or Target System. Tiles show Passed, Failed, and Warnings counts. Selecting a system tile opens run history for the related integrations.

**Screenshot needed: Monitoring grouped by Source System.**

## Run History and Message Processing Logs

The Monitoring Detail page displays Message Processing Logs. It filters out Discarded entries, formats SAP OData dates, and includes a Results column. When a captured payload is linked to a log row by Message ID, users can select View Results to preview or download the payload summary.

**Screenshot needed: Monitoring Detail page with Results link and payload dialog.**

# Standard Operational Workflows

## A. Find an integration by Source or Target System

Purpose: Identify which integration supports a vendor, HR platform, payroll system, time system, or third-party service.

1. Select Integrations.
2. Choose Sort by Source System or Sort by Target System.
3. Use Search integrations if the list is large.
4. Select the system tile.
5. Review the integration cards or list for that system.

Expected result: The page displays deployed integrations associated with the selected system.

Technical note: Source and Target grouping is based on Sender and Receiver metadata from design-time artifacts where available, with cached metadata in local storage.

## B. Review deployed integration details

1. Open Integrations.
2. Select the relevant Source or Target System.
3. Select an integration card.
4. Review artifact metadata, version, package, status, and externalized parameters.

Expected result: The Integration Detail view opens for the selected runtime artifact.

## C. Review externalized parameters

1. Open the integration detail page.
2. Expand the parameter group if needed.
3. Review the displayed key, label, and current value.

## D. Edit a supported configuration value

1. Open an integration detail page.
2. Change the required value.
3. Confirm that the Save draft or Deploy controls become relevant to the workflow.
4. Review the change before saving or deploying.

Expected result: The value is collected for a configuration update request.

## E. Validate configuration before redeployment

The code validates dirty state and collects parameter values, but it does not implement a full business validation rules engine. Integration owners must confirm values against the expected iFlow configuration rules before redeployment.

## F. Redeploy an integration

1. Open the integration detail page.
2. Review changed parameters.
3. Select Deploy.
4. Confirm the deployment dialog.

5. Wait for confirmation or error feedback.

Caution: Redeployment affects a tenant runtime artifact. Use change control appropriate for the environment.

## **G. Review or update a user-friendly schedule**

1. Open an integration with a timer-like parameter.
2. Locate the schedule section.
3. Choose Run Once, Schedule on Day, Schedule to Recur, or Advanced.
4. Update time, frequency, day, month, year, and time zone fields as required.
5. Review the generated schedule value before deployment.

## **H. Understand schedule mapping**

Timer-like parameters are detected by key or label text containing timer, cron, schedule, or frequency. The UI maps schedule choices into cron-like expression strings, including a --tz= suffix when a time zone is selected.

## **I. Monitor health by Source System**

1. Select Monitoring.
2. Choose Sort by Source System.
3. Review Passed, Failed, and Warnings counts on each system tile.
4. Select a tile to open run history.

## **J. Monitor health by Target System**

1. Select Monitoring.
2. Choose Sort by Target System.
3. Review the target system tiles.
4. Select a tile to investigate logs.

## **K. Drill into failed or warning runs**

1. Open Monitoring.
2. Select a system with failed or warning count.
3. Review Message Processing Logs.
4. Select a failed row to view error details when present.

## **L. Review message processing logs**

The log table shows Status, Integration, Message ID, Results, Sender, Time, and Duration. Discarded entries are filtered out by the application.

## **M. View a captured payload summary**

1. Open Monitoring Detail.
2. Find a log row with View Results.
3. Select View Results.
4. Review the preview in the payload dialog.

## **N. Download a downloadOnly payload**

1. Select View Results for the row.
2. If the payload is too large to preview, read the download-only message.
3. Select Download.
4. Handle the downloaded file according to HR data handling policy.

## **O. Escalate a failure**

1. Capture integration name, system, Message ID, time, status, error details.
2. Confirm whether a payload summary is available.
3. Escalate to Integration Support or the integration owner.
4. Do not attach prohibited PII to tickets.

## Monitoring And Troubleshooting Guide

| Symptom                             | Likely cause                                                                           | Where to investigate                                            | Recommended action                                          | Escalation owner      |
|-------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------|-----------------------|
| Integration missing from catalog    | Runtime artifact is undeployed, draft, or missing from IntegrationRuntimeArtifacts     | Integrations page and BTP Integration Suite                     | Confirm deployment status and runtime artifact availability | Integration Support   |
| No monitoring data displayed        | Destination/proxy unavailable, mock/live mode mismatch, or API error                   | Browser console, BackendClient mode, /api/v1 or /api/monitoring | Verify runtime mode and route mapping                       | SAP BTP Administrator |
| Source/Target grouping unexpected   | Sender/Receiver metadata missing or stale cached metadata                              | IntegrationDesignTimeArtifacts metadata and local storage cache | Refresh and validate design-time metadata                   | Integration Support   |
| Failed or warning totals unexpected | Runtime status or Message Processing Log aggregation differs from business expectation | Monitoring controller logic and MessageProcessingLogs           | Compare logs to Integration Suite                           | Integration Support   |
| Message logs unavailable            | MessageProcessingLogs API not reachable or filter mismatch                             | Monitoring detail, Integration Suite logs                       | Confirm integration ID and API permissions                  | Integration Developer |
| Payload Results link missing        | No payload with matching messageId was captured                                        | Payload API, PostgreSQL storage, iFlow HTTP call                | Send messageId from SAP_MessageProcessingLogID              | Integration Developer |
| Payload preview unavailable         | Payload exceeds 100 KB or preview flag is false                                        | Payload dialog and payload summary                              | Download file if permitted                                  | HRIS / Support        |

| Symptom                     | Likely cause                                                         | Where to investigate                            | Recommended action                      | Escalation owner      |
|-----------------------------|----------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------|-----------------------|
| Payload no longer available | Retention expired or database cleanup ran                            | PostgreSQL payload table                        | Re-run integration if business-approved | Application Owner     |
| Configuration update fails  | BTP API error, invalid parameter, or batch failure                   | BackendClient batch response, Integration Suite | Validate parameter key/value and retry  | Integration Support   |
| Redeployment fails          | DeployIntegrationDesignTimeArtifact error or token/destination issue | Deploy response, backend logs                   | Check tenant API permissions            | SAP BTP Administrator |
| Schedule update fails       | Invalid schedule mapping or unsupported target parameter             | Integration Detail schedule field               | Validate generated cron-like value      | Integration Developer |
| User cannot edit            | Role model may be handled outside code or not implemented            | Identity provider, BTP, app deployment          | Confirm production authorization model  | Application Owner     |

### Practical escalation packet

When escalating an issue, include: environment, integration name, Source System, Target System, Message ID, time, status, error details, whether Results payload was present, and any recent configuration changes. Do not include detailed employee data or prohibited PII.

# Payload Capture, Privacy, And Retention

## Business explanation

Payload capture lets an Integration Suite flow send a run result summary to Integration Pulse. The purpose is support visibility: users can see what summary result was produced for a run without opening Integration Suite or searching external logs.

## Supported formats

The backend API accepts JSON, CSV, XML, and plain text payload bodies. It also supports the original JSON wrapper contract.

## Endpoint

```
POST /payload-api/v1/payloads?integrationId=<artifact-id>&messageId=<message-id>&fileName=<file-name>
Content-Type: application/json | text/csv | application/xml | text/plain
```

Metadata may also be sent as headers: X-Integration-Id, X-Message-Id, and X-File-Name.

## Message Processing Log association

The Monitoring Results column links payloads to Message Processing Logs by messageId. SAP Integration Suite flows should set this value from the Message Processing Log ID, for example by using the SAP\_MessageProcessingLogID header in a Groovy script.

## Preview and download behavior

Payloads up to 100 KB are available for preview in the payload dialog. Payloads larger than 100 KB are stored but marked downloadOnly, and users must download them if permitted.

## Retention

Payloads are retained for one week. The PostgreSQL storage layer prunes expired rows when payload storage functions run. No separate scheduled cleanup job is present in the current repository.

## Privacy policy

Integration owners must ensure payloads contain only integration-run summaries. Payloads must not include PII other than permitted User IDs. They must not include employee names, emails, addresses, national IDs, payroll amounts, benefit elections, bank data, health data, or full transactional files.

## Implementation versus responsibility

| Topic          | Current implementation                                   | Responsibility remaining                                        |
|----------------|----------------------------------------------------------|-----------------------------------------------------------------|
| Format support | Accepts text payloads including JSON, CSV, XML, and text | Integration owner chooses appropriate summary format            |
| PII control    | No automatic PII detection or redaction verified         | Integration owner must prevent prohibited data from being sent  |
| Retention      | Seven-day expiration and pruning on storage access       | Platform owner must confirm operational cleanup expectations    |
| Access control | No role-based payload restriction verified in code       | Application owner must implement production authorization model |

## Allowed payload example

```
{
  "integrationName": "SuccessFactors to Payroll",
  "status": "Success",
}
```

```
"recordsProcessed": 125,  
"userIdsAffected": ["U123456", "U123457"],  
"executionTimestamp": "2026-06-22T14:30:00Z"  
}
```

## Prohibited content examples

- Employee names, personal emails, addresses, national IDs, bank details, compensation details, or benefit elections.
- Full payroll, benefits, or HR master-data extracts.
- Raw source or target payloads containing detailed employee data.

## Data lifecycle

See `assets/diagrams/payload_lifecycle.mmd` for the visual source.



# Administration And Configuration

## Integration artifact discovery

The application lists deployed/runtime integrations through `IntegrationRuntimeArtifacts` in destination mode or through `/api/integrations` in proxy mode. The frontend filters out statuses such as undeployed, not deployed, and draft.

## Parameter metadata

Externalized parameters are loaded from `IntegrationDesignTimeArtifacts(...)/Configurations` in destination mode or `/api/integrations/{id}/configurations` in proxy mode. Parameters are grouped by naming prefix, including Extract, Filter, Include, Delivery, Audit, SFTP, and General.

## Editable and read-only fields

The configuration model includes

read-only and secure fields, but the current documentation cannot confirm complete UI enforcement for every data source. Treat this as an area for validation before production.

## Parameter validation

The code tracks dirty state and collects parameter values. It does not implement full business validation for every parameter type. Integration owners remain responsible for validating tenant-specific parameter rules before redeployment.

## Redeployment workflow

Deployment first persists configuration values, then calls deployment. Destination mode uses a `$batch` request for configuration updates and then calls `DeployIntegrationDesignTimeArtifact`. The user receives confirmation and success/error feedback.

## Schedule transformation

Timer-like parameters are detected by text pattern and represented in a schedule UI. The selected schedule is transformed to a cron-like string. Time zone selections are appended using `--tz=<time-zone>`.

## Safe rollback considerations

The current code does not implement rollback or version restore. Before changing production parameters, capture current values, follow change-control procedures, and confirm a manual rollback path in SAP Integration Suite.

# Technical Architecture

## Logical architecture based on current repository evidence

The implemented repository contains:

- SAPUI5 XML/MVC frontend under webapp/.
- OpenUI5 dependencies declared in manifest.json and ui5.yaml.
- Runtime client logic in webapp/service/BackendClient.js.
- Optional FastAPI backend under backend/.
- SAP BTP Integration Suite API client placeholder/live implementation in backend/btp\_client.py.
- PostgreSQL-backed payload storage in backend/payload\_storage.py.
- Payload API routes in backend/routers/payloads.py.
- UI5 dev-server destination proxy mapping /api to a destination named Integration-Suite-Dev in ui5.yaml.

No MTA descriptor, approuter configuration, XSUAA descriptor, CI/CD pipeline, or HTML5 Applications Repository deployment descriptor is present in the repository.

## Component responsibilities

| Component               | Responsibility                                                     | Authentication evidence                         | Data handled                                              |
|-------------------------|--------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------|
| Browser/SAPUI5 frontend | User interface, routing, grouping, scheduling UI, monitoring views | No frontend auth enforcement verified           | Integration metadata, parameters, logs, payload summaries |
| BTP destination route   | Same-origin routing to Integration Suite API in destination mode   | Destination configuration external to repo      | Integration Suite OData requests                          |
| FastAPI proxy           | Optional proxy, health, CORS, CSP, BTP client, payload API         | OAuth client credentials in backend auth module | Integration data, logs, payload summaries                 |
| SAP Integration Suite   | Runtime artifacts, design-time configs, Message Processing Logs    | External to repo                                | Integration runtime and design-time data                  |
| PostgreSQL              | Payload storage                                                    | Service binding or DSN env var                  | Payload summary text, metadata, expiration                |

## Diagram source

|                                                                  |                                                                             |
|------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Browser / SAPUI5</b><br>Integration Pulse UI                  | <b>SAP BTP Integration Suite</b><br>Runtime artifacts, configurations, MPLs |
| <b>Optional FastAPI Proxy</b><br>OAuth token handling, CORS, CSP | <b>PostgreSQL</b><br>Payload summaries, seven-day retention                 |

Figure: Logical architecture based on current repository evidence. Mermaid source is included in assets/diagrams.

See assets/diagrams/logical\_architecture.mmd for Mermaid source.

# API And Backend Reference

## Frontend destination-mode routes

| Capability                 | Route called by frontend                                    | SAP API target                      |
|----------------------------|-------------------------------------------------------------|-------------------------------------|
| List integrations          | /api/v1/IntegrationRuntimeArtifacts                         | IntegrationRuntimeArtifacts         |
| Design-time metadata       | /api/v1/IntegrationDesigntimeArtifacts(Id=..., Version=...) | IntegrationDesigntimeArtifacts      |
| Configurations             | /api/v1/IntegrationDesigntimeArtifacts(...)/Configurations  | Configurations                      |
| Batch configuration update | /api/v1/                                                    | OData batch                         |
| Deployment                 | /api/v1/DeployIntegrationDesigntimeArtifact? ...            | DeployIntegrationDesigntimeArtifact |
| Message logs               | /api/v1/MessageProcessingLogs?...                           | MessageProcessingLogs               |

## Optional FastAPI proxy routes

| Method | Route                                             | Purpose                     |
|--------|---------------------------------------------------|-----------------------------|
| GET    | /health                                           | Health and mock-mode status |
| GET    | /api/integrations                                 | List integrations           |
| GET    | /api/integrations/{integration_id}                | Get integration detail      |
| GET    | /api/integrations/{integration_id}/configurations | Get externalized parameters |
| PUT    | /api/integrations/{integration_id}/configurations | Update parameters           |
| POST   | /api/integrations/{integration_id}/deploy         | Deploy integration          |
| GET    | /api/monitoring                                   | List monitoring items       |

| Method | Route                                          | Purpose                             |
|--------|------------------------------------------------|-------------------------------------|
| GET    | /api/monitoring/{integration_id}               | Get monitoring item                 |
| GET    | /api/monitoring/{integration_id}/logs          | Get message logs                    |
| POST   | /payload-api/v1/payloads                       | Receive payload summary             |
| GET    | /payload-api/v1/payloads?integrationId=...     | List payload summaries              |
| GET    | /payload-api/v1/payloads/{payload_id}          | Get payload detail when previewable |
| GET    | /payload-api/v1/payloads/{payload_id}/download | Download raw payload text           |

## Payload submission request

Raw text body with query metadata:

` ext

POST /payload-api/v1/payloads?integrationId=ExampleIntegration&messageId=ExampleMessage&fileName=Results06222026.json

Content-Type: application/json

`

Or wrapper body:

```
`json
{
  "integrationId": "ExampleIntegration",
  "messageId": "ExampleMessage",
  "fileName": "Results06222026.json",
  "contentType": "application/json",
  "payload": "{\"status\": \"Success\"}"
}
```

## Error handling notes

The frontend shows toast or dialog errors for failed loading, save, deploy, and payload loading operations. The FastAPI runtime error handler returns HTTP 502 for runtime configuration failures such as missing backend configuration.

# Security, Privacy, And Operations

## Authentication model

The frontend repository does not show a complete production authentication flow. The optional backend includes OAuth client-credentials handling for SAP Integration Suite API calls and keeps client secrets server-side. SuccessFactors embedding is supported through configurable Content-Security-Policy frame-ancestors values.

## Authorization model

Read Only and Editable roles are planned but not implemented in the current codebase. Production deployments must implement access control before claiming role enforcement.

## Credential handling

The backend reads secrets from environment variables. .env.example documents configuration categories; actual .env values must not be committed. The browser does not receive the BTP OAuth client secret in the optional proxy architecture.

## Payload privacy

Integration Pulse must not be used to store detailed employee data, payroll data, benefit elections, addresses, emails, national IDs, or other PII beyond permitted User IDs. The current implementation does not verify, redact, or reject PII automatically.

## Download controls

Download is available through the payload API for stored payloads. No role-based download control is verified in code. Organizations must control access through the application hosting and authorization layer.

## Operational safeguards

- Validate route mappings before production use.
- Confirm /payload-api points to the payload receiver service, not the Integration Suite destination.
- Bind PostgreSQL or set the DSN environment variable for payload storage.
- Confirm retention expectations and backup requirements.
- Implement role enforcement before production use.
- Review logs for sensitive data exposure.

# Deployment, Support, And Operational Runbook

## Verified build and local run commands

| Area                | Command                                                          |
|---------------------|------------------------------------------------------------------|
| Frontend install    | <code>npm install</code>                                         |
| Frontend mock start | <code>npm start</code> or <code>npm run start:mock</code>        |
| Destination mode    | <code>npm run start:destination</code>                           |
| Proxy mode          | <code>npm run start:proxy</code>                                 |
| UI build            | <code>npm run build</code>                                       |
| Backend local run   | <code>uvicorn main:app --reload --port 8000</code> from backend/ |

## SAP BTP prerequisites to validate

- Destination routing for Integration Suite OData API if using destination mode.
- Backend deployment for FastAPI proxy and payload API if using proxy/payload features.
- PostgreSQL service binding or DSN environment variable.
- CORS and frame ancestor settings for the hosting domain.
- Authentication and authorization model.

## Production readiness checklist

- ☐ Confirm final hosting architecture.
- ☐ Confirm /api route behavior.
- ☐ Confirm /payload-api route behavior.
- ☐ Bind PostgreSQL and test payload write/read/download.
- ☐ Implement and test role enforcement.
- ☐ Validate BTP destination or OAuth credentials.
- ☐ Test configuration update and deployment in non-production.
- ☐ Validate schedule parameter behavior with representative timer integrations.
- ☐ Confirm privacy policy and payload summary templates.
- ☐ Capture final screenshots for this guide.

## Operational handover checklist

- ☐ Identify application owner.
- ☐ Identify integration support owner.
- ☐ Identify SAP BTP platform owner.
- ☐ Document support hours and escalation paths.
- ☐ Document backup and retention expectations.
- ☐ Document release and rollback process.
- ☐ Review open items before production go-live.

# FAQ And Glossary

## Frequently asked questions

### *What is an externalized parameter?*

An externalized parameter is a configurable value exposed by an Integration Suite artifact so it can be changed without editing the iFlow design directly.

### *What is a runtime artifact?*

A runtime artifact is a deployed integration artifact available in the Integration Suite runtime.

### *What is a Message Processing Log?*

A Message Processing Log, or MPL, is the Integration Suite execution record for an integration run.

### *What does warning mean?*

In the application, warnings may represent states such as retry, processing, starting, deploying, or stopped, depending on runtime and log information.

### *Why is my payload download-only?*

Payloads over 100 KB are marked downloadOnly and cannot be previewed inline.

### *Why is a payload no longer available?*

Payloads expire after one week. The storage layer prunes expired records when accessed.

### *Why can I see an integration but not edit it?*

The current codebase does not enforce Read Only versus Editable roles. If editing is restricted in an environment, that restriction likely comes from deployment or identity-provider configuration outside the repository.

### *How is a Source System different from a Target System?*

The Source System is the sender or originating system. The Target System is the receiver or destination system.

### *What data may be sent to Integration Pulse?*

Only run summary information. User IDs may be included when approved. Detailed PII and transactional HR, payroll, benefits, or identity data must not be sent.

## Glossary

| Term                  | Definition                                               |
|-----------------------|----------------------------------------------------------|
| SAP BTP               | SAP Business Technology Platform                         |
| SAP Integration Suite | SAP service used to design, deploy, and run integrations |
| iFlow                 | Integration flow in SAP Integration Suite                |
| MPL                   | Message Processing Log for an integration run            |
| Source System         | The sender or originating system                         |
| Target System         | The receiver or destination system                       |
| Payload summary       | A small run-result payload intended for support review   |

| Term         | Definition                                                       |
|--------------|------------------------------------------------------------------|
| downloadOnly | Payload state used when a file is too large for browser preview  |
| Destination  | SAP BTP configuration used to route calls to an external service |

| Term          | Definition                                          |
|---------------|-----------------------------------------------------|
| FastAPI proxy | Optional backend service included in the repository |



## Open Items And Planned Enhancements

| Item                      | Status            | Notes                                                                     |
|---------------------------|-------------------|---------------------------------------------------------------------------|
| Final role model          | Planned           | Read Only and Editable are documented as planned, not enforced            |
| Future role types         | Planned           | Payload Viewer, Deployment Operator, and Administrator may be considered  |
| Production authentication | Validation needed | No complete XSUAA/approuter/SSO configuration is present                  |
| Deployment architecture   | Validation needed | No MTA, approuter, HTML5 repo, or CI/CD descriptor is present             |
| Screenshot package        | Needed            | Real screenshots should be generated from a sanitized running environment |
| Payload PII enforcement   | Not implemented   | No automatic PII detection or redaction is verified                       |
| Retention cleanup job     | Not present       | Cleanup occurs during storage access, not through a scheduled job in repo |

| Item                                          | Status            | Notes                                                                                   |
|-----------------------------------------------|-------------------|-----------------------------------------------------------------------------------------|
| Monitoring message counts in destination mode | Limited           | Destination mode currently sets messages24h/errors24h to zero unless enriched elsewhere |
| Read-only/secure parameter UI behavior        | Validation needed | Data model contains fields, but full enforcement needs confirmation                     |
| Production support model                      | Needed            | Owners, escalation paths, and SLAs must be defined                                      |

### Missing screenshots

- Home page with sidebar.
- Integrations page grouped by Source System.
- Integration Detail externalized parameters.
- Schedule UI for timer parameters.
- Monitoring page grouped by Source System.
- Monitoring Detail with Results column.
- Payload preview dialog and download-only message.